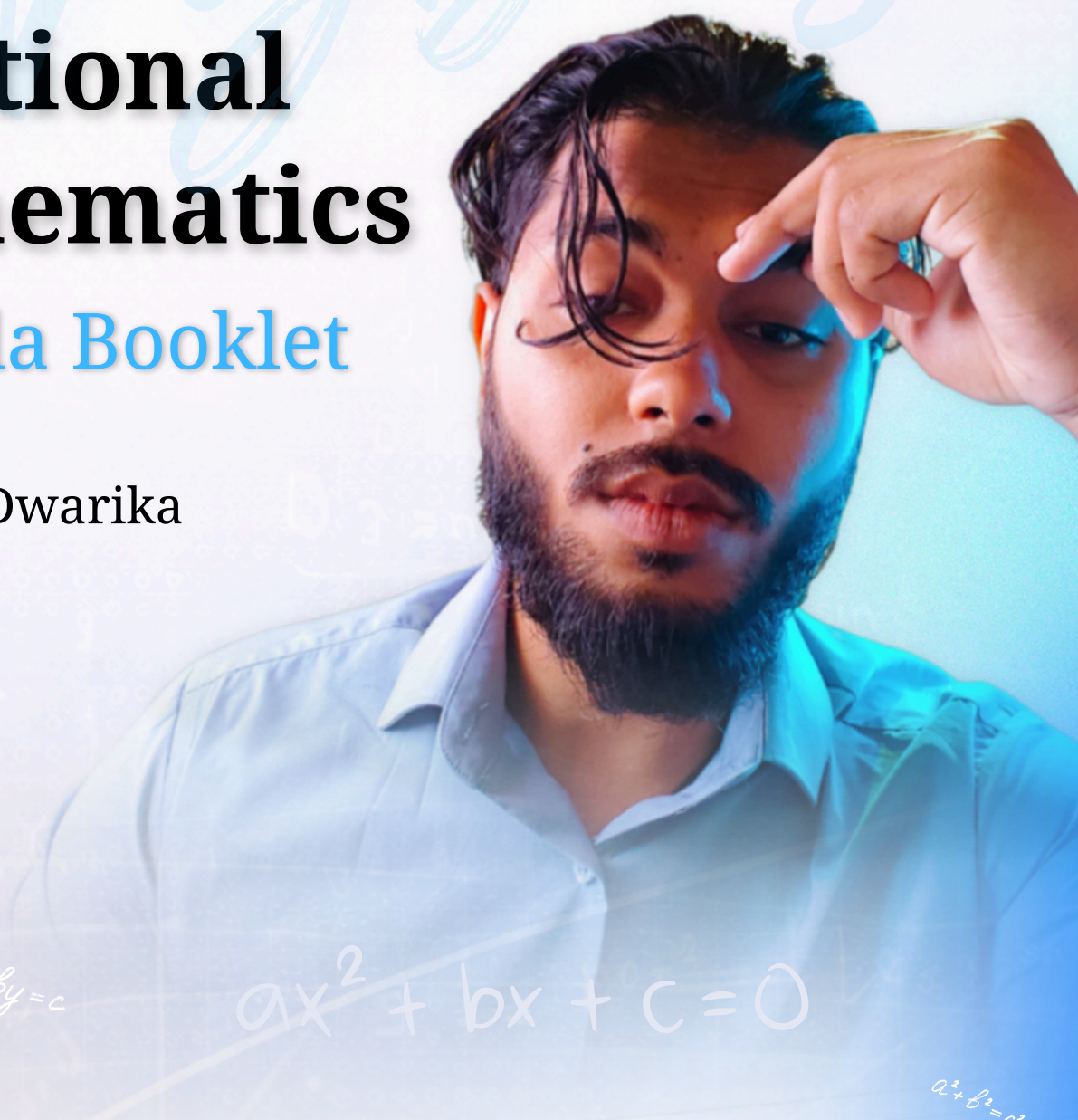


MATH

Vybes

Ultimate CSEC Additional Mathematics Formula Booklet

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Making Learning a *Vybe!*

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$y - y_1 = m(x - x_1)$$

$$a^2 + b^2 = c^2$$

$$ax^2 + bx + c = 0$$

$$ax + by = c$$

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Key Formulae

Allyuh better know this for CSEC Additional Mathematics

SECTION 1: Algebra, Sequences & Series

Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$D = b^2 - 4ac$ ($D > 0$: 2 real; $D = 0$: repeated; $D < 0$: complex)
Sum / Product of Roots	$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$
Completed Square	$f(x) = a(x + h)^2 + k, \quad h = \frac{b}{2a}, \quad k = c - \frac{b^2}{4a}$
Factor Theorem	$f(a) = 0 \Rightarrow (x - a)$ is a factor
Remainder Theorem	$f(a) = r$ when $f(x)$ is divided by $(x - a)$
Log-Index	$\log_a b = c \Leftrightarrow a^c = b$
Log Laws	$\log mn = \log m + \log n; \log \frac{m}{n} = \log m - \log n; \log m^n = n \log m$
Solving $a^x = b$	$x = \frac{\log b}{\log a}$
Linear Reduction	$y = ax^n \Rightarrow \log y = \log a + n \log x \quad y = ab^x \Rightarrow \log y = \log a + x \log b$
AP: nth term	$T_n = a + (n - 1)d$
AP: Sum	$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + T_n)$
GP: nth term	$T_n = ar^{n-1}$
GP: Sum	$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad \frac{a(1 - r^n)}{1 - r}$
GP: Sum to ∞	$S_\infty = \frac{a}{1 - r}, \quad r < 1$
Compound Interest	$A = P\left(1 + \frac{r}{100}\right)^T; \quad T = \frac{\log(A/P)}{\log(1 + r/100)}$

SECTION 2: Coordinate Geometry, Vectors & Trigonometry

Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Line forms	$y = mx + c$; $y - y_1 = m(x - x_1)$; $ax + by + c = 0$
Parallel / Perp.	Parallel: $m_1 = m_2$; Perpendicular: $m_1 m_2 = -1$
Circle (standard)	$(x + f)^2 + (y + g)^2 = r^2$; centre $(-f, -g)$, radius r
Circle (general)	$x^2 + y^2 + 2fx + 2gy + c = 0$; $r = \sqrt{f^2 + g^2 - c}$
Midpoint / Distance	$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$; $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Vector magnitude	$ \mathbf{v} = \sqrt{x^2 + y^2}$
Unit vector	$\hat{\mathbf{v}} = \mathbf{v}/ \mathbf{v} $
Dot product	$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 = \mathbf{a} \mathbf{b} \cos \theta$
Angle between vectors	$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} \mathbf{b} }$
Perp. vectors	$\mathbf{a} \cdot \mathbf{b} = 0$
Arc length / Sector	$\ell = r\theta$; $A = \frac{1}{2}r^2\theta$ (θ in radians)
Triangle area	$\text{Area} = \frac{1}{2}ab \sin C$
Pythagorean identity	$\sin^2 \theta + \cos^2 \theta = 1$; $\tan \theta = \sin \theta / \cos \theta$
Compound angles	$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
Double angles	$\sin 2A = 2 \sin A \cos A$; $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$ $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

SECTION 3: Introductory Calculus

Power rule (diff.)	$\frac{d}{dx}(x^n) = nx^{n-1}$
Trig derivatives	$\frac{d}{dx} \sin ax = a \cos ax$; $\frac{d}{dx} \cos ax = -a \sin ax$; $\frac{d}{dx} \tan x = \sec^2 x$
Product rule	$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$
Quotient rule	$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$
Chain rule	$\frac{d}{dx}[f(ax+b)] = a \cdot f'(ax+b)$
First principles	$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
Stationary point	$\frac{dy}{dx} = 0$; $f'' > 0 \Rightarrow \min$, $f'' < 0 \Rightarrow \max$
Tangent / Normal	Tangent gradient = $f'(a)$; Normal gradient = $-1/f'(a)$; use $y - y_1 = m(x - x_1)$
Power rule (int.)	$\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$
Trig integrals	$\int \sin ax dx = -\frac{1}{a} \cos ax + c$; $\int \cos ax dx = \frac{1}{a} \sin ax + c$
Definite integral	$\int_a^b f(x) dx = F(b) - F(a)$
Area under curve	$A = \int_a^b f(x) dx$
Volume of revolution	$V = \pi \int_a^b [f(x)]^2 dx$
Curve from gradient	Integrate dy/dx , substitute known point to find c
Kinematics	$v = \frac{ds}{dt}$; $a = \frac{dv}{dt}$; $s = \int v dt$; $v = \int a dt$
SUVAT	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$

SECTION 4: Probability & Statistics

Mean (ungrouped)	$\bar{x} = \frac{\sum x_i}{n}$
Mean (grouped)	$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
Quartile positions	$Q_1 = \frac{1}{4}(n+1)\text{th term} ; \quad Q_2 = \frac{1}{2}(n+1)\text{th term} ; \quad Q_3 = \frac{3}{4}(n+1)\text{th term}$
IQR / Semi-IQR	$\text{IQR} = Q_3 - Q_1 ; \quad \text{Semi-IQR} = \frac{Q_3 - Q_1}{2}$
Variance (ungrouped)	$S^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x_i^2}{n} - \bar{x}^2$
Variance (grouped)	$S^2 = \frac{\sum f_i x_i^2}{\sum f_i} - \bar{x}^2$
Std. deviation	$S = \sqrt{S^2}$
Classical probability	$P(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$
Complement	$P(A') = 1 - P(A)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive	$P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B)$
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Independent events	$P(A \cap B) = P(A) \times P(B) ; \quad P(A B) = P(A)$

SECTION 1: ALGEBRA, SEQUENCES & SERIES

Quadratics

A **quadratic equation** is of the form:

$$ax^2 + bx + c = 0$$

where $a \neq 0$, and α and β are the roots (solutions) of the quadratic.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- What:** The **quadratic formula** — gives the exact solutions of a quadratic.
When: Use when the equation is not easily factorised or when asked for exact roots.
Why: Derived from completing the square, it works for all quadratics, real or complex.

$$\text{Discriminant: } D = b^2 - 4ac$$

- What:** Tells the nature of the roots of a quadratic equation.
When: Use before solving to determine if the roots are real or complex.
Why: Because:
 - $D > 0$: Two distinct real roots
 - $D = 0$: One real repeated root
 - $D < 0$: Two complex (conjugate) roots

$$(x - \alpha)(x - \beta) = 0 \Rightarrow x = \alpha \text{ or } \beta$$

- What:** Factorised form of a quadratic equation.
When: Use when the quadratic can be factorised neatly.
Why: This form shows the solutions explicitly.
 Here, α and β are the roots of the quadratic.

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

- What:** The **sum and product of roots** formulas (Vieta's formulas).
When: Use in problems involving unknown roots, symmetric relationships, or root-based identities.
Why: Comes from expanding the factorised form:
 $a(x - \alpha)(x - \beta) = ax^2 - a(\alpha + \beta)x + a\alpha\beta$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

- What:** Rebuilding the quadratic from its roots.
When: Use when you're given the roots α and β , and asked to form the equation.
Why: This is the standard form derived from multiplying out $(x - \alpha)(x - \beta)$.

$$f(x) = a(x + h)^2 + k \quad \text{where } h = -\frac{b}{2a}, \quad k = c - \frac{b^2}{4a}$$

- What:** This is the **completed square form** of a quadratic.
When: Use when finding the **vertex (turning point)** or rewriting the function for graph sketching or transformations.
Why: This form makes the minimum or maximum value of the function clear.

Interpreting the formula:

- a : Controls the direction and width of the parabola (positive opens up, negative opens down).
- $h = -\frac{b}{2a}$: Gives the horizontal shift — note the sign is **opposite** inside the bracket.
- $k = c - \frac{b^2}{4a}$: Vertical shift (the actual minimum or maximum value of the function).

$$\text{Turning Point: } (-h, k)$$

- What:** The vertex (lowest or highest point) of the parabola.
When: Use when asked for the max/min value or to sketch the graph.
Why: The quadratic is symmetrical about this point, and the square term ensures it's the extremum.

$$\text{Axis of Symmetry: } x = -h = -\frac{b}{2a}$$

- What:** A vertical line that divides the parabola into two mirror-image halves.
When: Use when sketching or describing the symmetry of the graph.
Why: Because the turning point lies on this line, and the shape of a parabola is always symmetric.

If $f(\alpha) = 0$, then $(x - \alpha)$ is a factor of $f(x)$

- What:** This is the **Factor Theorem**.
When: Use when factorising a cubic or quadratic and testing possible roots.
Why: If substituting a value into a polynomial gives 0, then that value is a root and the corresponding binomial is a factor.

Factor Theorem Example

Let's say we have the polynomial:

$$f(x) = x^2 - 5x + 6$$

Suppose we want to check if $(x - 2)$ is a factor.

Step 1: Use the Factor Theorem Substitute $x = 2$ into the function:

$$f(2) = (2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$$

Conclusion: Since $f(2) = 0$, then by the **Factor Theorem**,

$(x - 2)$ is a factor of $f(x)$.

If $f(a) = r$, then r is the remainder when $f(x)$ is divided by $(x - a)$

- What:** This is the **Remainder Theorem**.
When: Use to find the remainder quickly without full division.
Why: Plugging a into the function mimics the division process — it gives the result directly.

Remainder Theorem Example

Now suppose we want the remainder when:

$$f(x) = 2x^3 + 3x^2 - x + 5$$

is divided by $x - 1$.

Step 1: Use the Remainder Theorem Substitute $x = 1$ into $f(x)$:

$$f(1) = 2(1)^3 + 3(1)^2 - 1 + 5 = 2 + 3 - 1 + 5 = 9$$

Conclusion: The remainder when $f(x)$ is divided by $x - 1$ is **9**.

Polynomial Long Division - Part of Quadratics

$$\begin{array}{r} x^2 + 4 \\ x - 2 \overline{) x^3 - 2x^2 + 4x - 8} \\ \underline{-x^3 + 2x^2} \\ 4x - 8 \\ \underline{-4x + 8} \\ 0 \end{array}$$

What: Polynomial long division is a method for dividing one polynomial by another.

When: Use when simplifying rational expressions, checking for factors, or applying the Factor/Remainder Theorems.

Why: This method expresses a polynomial as:

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

Conclusion:

$$\frac{x^3 - 2x^2 + 4x - 8}{x - 2} = x^2 + 4 \quad (\text{since remainder} = 0)$$

So, $x - 2$ is a factor of the cubic polynomial.

Equations Reducible to Quadratic

Some equations that look complicated can be solved by making a **substitution** to turn them into a quadratic.

$$\text{Let } u = f(x) \text{ to reduce to } au^2 + bu + c = 0$$

- What:** A technique for solving equations of higher degree or non-standard form.
When: Use when you spot a pattern like $(f(x))^2 + b(f(x)) + c = 0$.
Why: The substitution reduces the equation to a standard quadratic, which you solve by factorising or the formula — then back-substitute.

Example 1: $x^4 - 6x^2 + 8 = 0$

Let $u = x^2$:

$$u^2 - 6u + 8 = 0 \Rightarrow (u - 2)(u - 4) = 0 \Rightarrow u = 2 \text{ or } 4$$

$$x^2 = 2 \Rightarrow x = \pm\sqrt{2} \quad x^2 = 4 \Rightarrow x = \pm 2$$

Example 2: $x - 2\sqrt{x} + 1 = 0$

Let $u = \sqrt{x}$ so $u^2 = x$:

$$u^2 - 2u + 1 = 0 \Rightarrow (u - 1)^2 = 0 \Rightarrow u = 1 \Rightarrow x = 1$$

Simultaneous Equations — One Linear, One Non-Linear

When one equation is linear and the other is a quadratic or circle, **substitute** the linear expression into the non-linear one.

From linear equation: express y in terms of x . Substitute into non-linear equation.

- What:** Solving a system where a line meets a curve.
When: Use when asked to find intersection points of a line with a parabola or circle.
Why: Substitution reduces the system to a single quadratic. The discriminant tells you how many solutions (intersections) exist.

Example 1: Line and Quadratic

$y = x + 2$ and $y = x^2 - 4$: substitute $y = x + 2$:

$$x + 2 = x^2 - 4 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$$

$$x = 3, y = 5 \quad \text{or} \quad x = -2, y = 0$$

Example 2: Line and Circle

$y = 2x - 1$ and $x^2 + y^2 = 5$: substitute $y = 2x - 1$:

$$x^2 + (2x - 1)^2 = 5 \Rightarrow 5x^2 - 4x - 4 = 0$$

Solve using the quadratic formula and back-substitute.

Inequalities

Quadratic Inequalities

To solve a quadratic inequality, factorise, find the roots, then determine the sign of the expression in each region.

$$ax^2 + bx + c > 0 \quad \text{or} \quad ax^2 + bx + c < 0$$

- What:** Quadratic inequality — find the values of x that satisfy it.
When: Use when you need to solve expressions like $x^2 - 5x + 6 > 0$.
Why: The parabola changes sign at its roots, so testing regions tells you where the inequality holds.

Set-Builder Notation: Solutions to inequalities can be written as sets.

- $\{x : 2 < x < 3\}$ means “the set of all x such that x is between 2 and 3”
- $\{x : x < -1 \text{ or } x > 4\}$ for two separate regions

Steps to Solve a Quadratic Inequality

1. Rearrange so one side is 0.
2. Factorise and find roots α and β (assume $\alpha < \beta$).
3. Sketch the parabola (or use a sign diagram).
4. If $a > 0$: the parabola opens **upward**:

$$f(x) < 0 \Rightarrow \alpha < x < \beta \quad f(x) > 0 \Rightarrow x < \alpha \text{ or } x > \beta$$

5. If $a < 0$: flip the regions above.

Example: Solve $x^2 - 5x + 6 < 0$

Factorise: $(x - 2)(x - 3) < 0$. Roots are $x = 2$ and $x = 3$. Parabola opens up, so:

$$2 < x < 3$$

Rational Inequalities

$$\frac{ax + b}{cx + d} > 0 \quad (\text{or } \geq 0, < 0, \leq 0)$$

- What:** Inequality with a linear expression on top and bottom.
When: Use when you see a fraction with x in the denominator inside an inequality.
Why: You **cannot** multiply across by $(cx + d)$ without knowing its sign — use a sign diagram instead.

Method: Sign Diagram

1. Find the critical values: set numerator = 0 and denominator = 0 separately.
2. Place critical values on a number line.
3. Test the sign of the fraction in each region.
4. Select the regions that satisfy the inequality. Note: the value where the denominator is zero is **always excluded**.

Example: Solve $\frac{x-1}{x+3} > 0$

Critical values: $x = 1$ and $x = -3$.

Region	$x < -3$	$-3 < x < 1$	$x > 1$
Sign of fraction	$\frac{(-)}{(-)} = +$	$\frac{(-)}{(+)} = -$	$\frac{(+)}{(+)} = +$

$$x < -3 \quad \text{or} \quad x > 1$$

Surds

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

- What:** Multiply two surds.
When: Use when simplifying products like $\sqrt{2} \cdot \sqrt{3}$.
Why: The square root of a product is the product of the square roots.

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

- What:** Division of surds.
When: Use when simplifying a fraction of roots.
Why: The square root distributes over division, just like multiplication.

$$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$$

- What:** Rationalising a simple surd in the denominator.
When: Use when the denominator is a single square root.
Why: Multiplying numerator and denominator by $\sqrt{a}$ removes the surd below.

$$\frac{1}{a + \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$$

- What:** Rationalising a binomial denominator using the conjugate.
When: Use when the denominator has the form $a + \sqrt{b}$.
Why: $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$, which eliminates the surd.

Indices (Exponents)

$$a^m \times a^n = a^{m+n}$$

- What:** Multiply powers with the same base by adding their exponents.
When: Use when simplifying expressions like $2^3 \times 2^4$.
Why: You're combining the total number of times the base appears as a factor.

$$\frac{a^m}{a^n} = a^{m-n}$$

What: Divide powers with the same base by subtracting the exponents.
When: Use when simplifying expressions like $\frac{3^5}{3^2}$.
Why: You cancel common factors in numerator and denominator.

$$(a^m)^n = a^{mn}$$

What: Power of a power rule.
When: Use when an exponent is raised to another exponent.
Why: You're multiplying repeated groups of the same base.

$$a^0 = 1 \quad (a \neq 0)$$

What: Any non-zero number raised to zero equals one.
When: Use when exponents cancel each other.
Why: From $\frac{a^n}{a^n} = a^0 = 1$.

$$a^{-n} = \frac{1}{a^n}$$

What: A negative exponent represents the reciprocal.
When: To convert negative exponents into fractional form.
Why: Negative powers "flip" the base to the denominator.

$$\sqrt[n]{a} = a^{1/n}$$

What: Roots as fractional indices.
When: To simplify or apply exponent rules.
Why: $(a^{1/n})^n = a$, same as $\sqrt[n]{a}$.

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

What: Fractional indices combine roots and powers.
When: Use when evaluating or simplifying expressions like $8^{2/3}$.
Why: The denominator gives the root, and the numerator gives the power.

Logarithms

$$\log_b(mn) = \log_b m + \log_b n$$

- What:** Product Rule — the log of a product equals the sum of the logs.
When: Use when you're multiplying numbers inside a logarithm, like $\log_2(8 \times 4)$.
Why: This works because logarithms are exponents, and adding exponents corresponds to multiplying numbers.

$$\log_b\left(\frac{m}{n}\right) = \log_b m - \log_b n$$

- What:** Quotient Rule — the log of a division equals the difference of the logs.
When: Use when simplifying or expanding expressions like $\log_5\left(\frac{25}{5}\right)$.
Why: Subtracting exponents corresponds to division, since $a^m/a^n = a^{m-n}$.

$$\log_b(m^n) = n \log_b m$$

- What:** Power Rule — the exponent inside a log can be moved in front.
When: Use when evaluating or simplifying logs of powers, e.g. $\log_3(27^2)$.
Why: This reflects the way exponents and logs interact: if $b^x = m$, then $b^{nx} = m^n$, so $\log_b m^n = nx$.

$$\log_a b = \frac{\log_c b}{\log_c a}$$

- What:** Change of Base Formula.
When: Use when your calculator or exam only supports base 10 or base e logs.
Why: Allows you to rewrite any logarithm in terms of another base, using the definition of logs as exponents.

$$\log_b b = 1$$

- What:** The log of a number in its own base is 1.
When: Use when simplifying expressions like $\log_7 7$.
Why: Because $b^1 = b$, so the exponent (logarithm) must be 1.

$$\log_b 1 = 0$$

- What:** The log of 1 is always zero, regardless of the base (as long as $b > 0, b \neq 1$).
When: Use when simplifying or solving equations like $\log_2(x) = 0$.
Why: Because any number raised to the power 0 is 1.

Log-Index Conversion

$$\log_a b = c \iff a^c = b$$

- What:** The definition of a logarithm — it is the **exponent** you raise the base to in order to get b .
When: Use to convert between log form and index (exponential) form, which is the first step in solving most log equations.
Why: Logarithms *are* exponents. This equivalence is the bridge between the two notations.
Example: $\log_2 8 = 3 \Leftrightarrow 2^3 = 8$

Solving Exponential Equations: $a^x = b$

$$a^x = b \implies x = \frac{\log b}{\log a}$$

- What:** Method for solving equations where the unknown is in the exponent.
When: Use when you cannot express both sides with the same base — take log of both sides.
Why: Taking logs brings the exponent x down using the power rule: $x \log a = \log b$.
Example: $3^x = 20 \Rightarrow x = \frac{\log 20}{\log 3} \approx 2.73$

Linear Reduction (Log-Linear Form)

Some relationships that are not linear can be **converted to linear form** using logarithms. This allows you to identify the equation of the curve from a straight-line graph.

Type 1: Power Law — $y = ax^n$

$$y = ax^n \implies \log y = \log a + n \log x$$

What: Transforms a power relationship into $Y = mX + c$ form.
When: Use when the graph of $\log y$ vs $\log x$ is a straight line.
Why: Taking logs of both sides brings the exponent down, producing a linear equation.
 Plot $\log y$ on the y -axis and $\log x$ on the x -axis:
Gradient = n , **Intercept** = $\log a$

Type 2: Exponential Law — $y = ab^x$

$$y = ab^x \implies \log y = \log a + x \log b$$

What: Transforms an exponential relationship into $Y = mX + c$ form.
When: Use when the graph of $\log y$ vs x is a straight line.
Why: The exponent x comes out front, making the equation linear in x .
 Plot $\log y$ on the y -axis and x on the x -axis:
Gradient = $\log b$, **Intercept** = $\log a$

Quick Reference Table

Relationship	Linear Form	Plot	Gradient	Intercept
$y = ax^n$	$\log y = \log a + n \log x$	$\log y$ vs $\log x$	n	$\log a$
$y = ab^x$	$\log y = \log a + x \log b$	$\log y$ vs x	$\log b$	$\log a$

Sequences, Series, and Summation — What, When, and Why

Arithmetic Sequences (A.P.)

A sequence is arithmetic if it increases or decreases by a **common difference** d each time.

$$T_n = a + (n - 1)d$$

- What:** The n th term of an arithmetic sequence.
When: Use to find any term when the first term a and common difference d are known.
Why: The formula adds the common difference $(n - 1)$ times to the first term.

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

Or alternatively:

$$S_n = \frac{n}{2}(a + T_n)$$

- What:** The sum of the first n terms of an arithmetic sequence.
When: Use to calculate the total value of a fixed number of terms.
Why: It averages the first and last terms, then multiplies by the number of terms.

Geometric Sequences (G.P.)

A sequence is geometric if each term is found by multiplying the previous term by a constant called the **common ratio** r .

$$T_n = ar^{n-1}$$

- What:** The n th term of a geometric sequence.
When: Use to find any term when the first term a and common ratio r are known.
Why: The formula multiplies the starting term by the common ratio r , $(n - 1)$ times.

$$S_n = \frac{a(r^n - 1)}{r - 1}, \quad \text{when } r > 1$$

$$S_n = \frac{a(1 - r^n)}{1 - r}, \quad \text{when } 0 < r < 1$$

- What:** The sum of the first n terms of a geometric sequence.
- When:** Use to calculate the total value of a geometric series. Choose the correct form based on the value of r .
- Why:** The formula is derived by subtracting the series from a scaled version of itself.

$$S_{\infty} = \frac{a}{1-r}, \quad \text{if } |r| < 1$$

- What:** The sum to infinity of a geometric sequence.
- When:** Use only when the common ratio $|r| < 1$, and the terms get smaller.
- Why:** As $n \rightarrow \infty$, $r^n \rightarrow 0$, so the finite sum becomes exact.

Convergence/Divergence:

Geometric series converges (has a finite sum to ∞) if and only if $|r| < 1$

Geometric series diverges (no finite sum) if $|r| \geq 1$

All arithmetic series diverge (except the trivial case $d = 0$)

- What:** Rules for whether an infinite series approaches a finite value.
- When:** Always check $|r|$ before using S_{∞} . If $|r| \geq 1$, S_{∞} does not exist.
- Why:** If $|r| \geq 1$, terms don't shrink — the series grows without bound. If $|r| < 1$, terms shrink to zero, allowing a finite total.

Summation Notation $\sum$

The **Greek letter sigma** $\sum$ is used to represent a **sum of terms** in a sequence.

$$\sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

- What:** Summation notation tells you to add up a sequence of terms from $k = 1$ to $k = n$.
- When:** Use when a sequence is written compactly in sigma form, especially in exam questions involving patterns or series.
- Why:** It simplifies notation and clearly defines the number of terms to add.

Parts of the Sigma Notation:

Symbol	Meaning
$\sum$	"Sum of"
k	Index of summation — starts at the lower limit and increases by 1
n	Upper limit — the final value of the index
a_k	General term — the expression that defines the k th term

Connections to A.P. and G.P.

- If $a_k = a + (k - 1)d$, then:

$$\sum_{k=1}^n a_k = S_n = \frac{n}{2}[2a + (n - 1)d]$$

- If $a_k = ar^{k-1}$, then:

$$\sum_{k=1}^n a_k = S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad \frac{a(1 - r^n)}{1 - r}$$

Example:

$$\sum_{k=1}^4 (3 + 2(k - 1)) = 3 + 5 + 7 + 9 = 24$$

- This is an **A.P.** with $a = 3$, $d = 2$, $n = 4$ - Use:

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{4}{2}[6 + 6] = 24$$

Example (G.P. using $\sum$ notation):

$$\sum_{k=1}^4 \left(2 \cdot \left(\frac{1}{2} \right)^{k-1} \right) = 2 + 1 + 0.5 + 0.25 = 3.75$$

- This is a **geometric series** with: - First term: $a = 2$ - Common ratio: $r = \frac{1}{2}$ - Number of terms: $n = 4$

Using the G.P. Sum Formula:

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_4 = \frac{2(1 - (1/2)^4)}{1 - 1/2}$$

$$S_4 = \frac{2(1 - 0.0625)}{0.5}$$

$$S_4 = \frac{2 \cdot 0.9375}{0.5}$$

$$S_4 = \frac{1.875}{0.5}$$

$$S_4 = 3.75$$

What:

G.P. partial sum using sigma notation.

When:

Use when the number of terms is known and you're summing a geometric series.

Why:

The formula subtracts a scaled version of the sequence from itself and solves algebraically.

Compound Interest and Logarithmic Rearrangement

Compound Interest Formula

$$A = P \left(1 + \frac{r}{100} \right)^T$$

- What:** The compound interest formula — calculates the **future value** of an investment or salary over time with a fixed percentage increase.
- When:** Use when a quantity grows (or shrinks) **exponentially** over regular intervals — e.g., savings, loans, salaries.
- Why:** Because each period earns interest **on top of the previous amount**, not just on the original principal.

Example:

Problem: You invest \$4000 at an interest rate of 8% per year. How much will it grow to after 5 years?

Step 1: Identify variables - $P = 4000$, $r = 8$, $T = 5$

Step 2: Apply the formula

$$A = 4000 \left(1 + \frac{8}{100} \right)^5$$

$$A = 4000(1.08)^5$$

$$A = 4000 \cdot 1.4693$$

$$A = 5877.20$$

Conclusion: The final amount after 5 years is **\$5877.20**.

Solving for Time Using Logarithms

This is useful when you're **given** A , P , **and** r , and asked **how long** until an amount is reached — e.g., **2021 Q2C**.

$$T = \frac{\log \left(\frac{A}{P} \right)}{\log \left(1 + \frac{r}{100} \right)}$$

- What:** Rearranged version of the compound interest formula to solve for time.
- When:** Use when the final amount is known, and you need to find the time it takes to reach that amount.
- Why:** You apply logarithms because the exponent contains the unknown variable T , and logs are the inverse of exponentiation.

Step-by-Step Rearrangement

Start from:

$$A = P \left(1 + \frac{r}{100}\right)^T$$

Step 1: Divide both sides by P

$$\frac{A}{P} = \left(1 + \frac{r}{100}\right)^T$$

Step 2: Take logs of both sides

$$\log\left(\frac{A}{P}\right) = \log\left[\left(1 + \frac{r}{100}\right)^T\right]$$

Step 3: Use the power rule of logs

$$\log\left(\frac{A}{P}\right) = T \cdot \log\left(1 + \frac{r}{100}\right)$$

Step 4: Solve for T

$$T = \frac{\log\left(\frac{A}{P}\right)}{\log\left(1 + \frac{r}{100}\right)}$$

Example:

Problem: How long will it take for an investment of \$4000 to triple at 8% interest?

Step 1: Identify variables - $P = 4000$, $A = 12000$, $r = 8$

Step 2: Apply the formula

$$T = \frac{\log\left(\frac{12000}{4000}\right)}{\log(1.08)}$$

$$T = \frac{\log(3)}{\log(1.08)}$$

$$T = \frac{0.4771}{0.0334}$$

$$T \approx 14.28$$

Conclusion: It takes about 15 years to triple the investment.

Common Strategies for Solving Sequence, Series, and Summation Questions

These strategies help you **recognize**, **plan**, and **solve** problems efficiently — especially under exam pressure.

Strategy 1: Identify the Type of Sequence First

If the terms increase or decrease by a **fixed amount** → **Arithmetic**

If the terms are multiplied or divided by a **fixed ratio** → **Geometric**

If the pattern is not obvious → write out terms and check differences or ratios

- What:** Determine whether the sequence is arithmetic, geometric, or neither.
When: Use at the start of any sequence or series problem to decide which formulas to apply.
Why: Correctly identifying the sequence type ensures you use the right formulas and avoid mistakes.

Strategy 2: Use Given Terms to Set Up Equations

Given terms can help you find unknowns like a , d , or r .

- What:** Use known terms to create equations and solve for unknowns.
When: Use when you're given specific terms in a sequence (e.g., $T_3 = 25$, $T_1 + T_2 = 150$).
Why: This allows you to find the first term a , common difference d , or common ratio r .

Example:

Problem: You're told: - $T_3 = 25$ - $T_1 + T_2 = 150$

If it's a G.P., use:

$$ar^2 = 25 \quad a + ar = 150$$

Step 1: Solve for a and r From $a + ar = 150$, factor out a :

$$a(1 + r) = 150$$

From $ar^2 = 25$, solve for a :

$$a = \frac{25}{r^2}$$

Step 2: Substitute a into the first equation

$$\frac{25}{r^2}(1 + r) = 150$$

$$25(1 + r) = 150r^2$$

$$25 + 25r = 150r^2$$

$$150r^2 - 25r - 25 = 0$$

Step 3: Solve the quadratic equation Using the quadratic formula:

$$r = \frac{25 \pm \sqrt{(-25)^2 - 4 \cdot 150 \cdot (-25)}}{2 \cdot 150}$$

$$r = \frac{25 \pm \sqrt{625 + 15000}}{300}$$

$$r = \frac{25 \pm \sqrt{15625}}{300}$$

$$r = \frac{25 \pm 125}{300}$$

Step 4: Find r and a

$$r = \frac{150}{300} = 0.5 \quad \text{or} \quad r = \frac{-100}{300} = -\frac{1}{3}$$

If $r = 0.5$:

$$a = \frac{25}{(0.5)^2} = \frac{25}{0.25} = 100$$

If $r = -\frac{1}{3}$:

$$a = \frac{25}{\left(-\frac{1}{3}\right)^2} = \frac{25}{\frac{1}{9}} = 225$$

Conclusion: The sequence could have: - $a = 100$, $r = 0.5$ - $a = 225$, $r = -\frac{1}{3}$

Strategy 3: Split the Series into Two When Necessary

If a series alternates or has a complex pattern, split it into simpler parts.

What: Break a complex series into two or more simpler series.

When: Use when the series alternates or has a pattern that can be separated (e.g., odd and even terms).

Why: Simplifying the series makes it easier to apply formulas for arithmetic or geometric sequences.

Example:

Problem:

$$\frac{1}{4} + \frac{2}{4^2} + \frac{1}{4^3} + \frac{2}{4^4} + \dots$$

Step 1: Split the series - Odd terms: $\frac{1}{4} + \frac{1}{4^3} + \dots$ - Even terms: $\frac{2}{4^2} + \frac{2}{4^4} + \dots$

Step 2: Identify the type of each series - Odd terms: Geometric with $a = \frac{1}{4}$, $r = \frac{1}{16}$
 - Even terms: Geometric with $a = \frac{2}{16} = \frac{1}{8}$, $r = \frac{1}{16}$

Step 3: Apply the sum formula for each series For odd terms:

$$S_{\text{odd}} = \frac{\frac{1}{4}}{1 - \frac{1}{16}} = \frac{\frac{1}{4}}{\frac{15}{16}} = \frac{16}{60} = \frac{4}{15}$$

For even terms:

$$S_{\text{even}} = \frac{\frac{1}{8}}{1 - \frac{1}{16}} = \frac{\frac{1}{8}}{\frac{15}{16}} = \frac{16}{120} = \frac{2}{15}$$

Step 4: Add the sums

$$S_{\text{total}} = S_{\text{odd}} + S_{\text{even}} = \frac{4}{15} + \frac{2}{15} = \frac{6}{15} = \frac{2}{5}$$

Conclusion: The sum of the series is $\frac{2}{5}$.

Strategy 4: Choose the Correct Sum Formula for G.P.

$$\text{Use } S_n = \frac{a(r^n - 1)}{r - 1} \text{ when } r > 1$$

$$\text{Use } S_n = \frac{a(1 - r^n)}{1 - r} \text{ when } 0 < r < 1$$

$$\text{Use } S_{\infty} = \frac{a}{1 - r} \text{ when } |r| < 1$$

What: Select the appropriate sum formula based on the value of the common ratio r .

When: Use when calculating the sum of a geometric series.

Why: The formulas are derived differently depending on whether $r > 1$, $0 < r < 1$, or $|r| < 1$.

Example:

Problem: Find the sum of the first 5 terms of the geometric sequence with $a = 3$ and $r = 2$.

Step 1: Identify the formula Since $r = 2 > 1$, use:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Step 2: Plug in the values

$$S_5 = \frac{3(2^5 - 1)}{2 - 1} = \frac{3(32 - 1)}{1} = \frac{3 \cdot 31}{1} = 93$$

Conclusion: The sum of the first 5 terms is **93**.

Strategy 5: Recognize Infinite Series Conditions

If $|r| < 1$, the series converges and has a sum: $S_{\infty} = \frac{a}{1-r}$

If $|r| \geq 1$, the series diverges and has **no finite sum**

- What:** Determine whether a geometric series converges or diverges.
When: Use when dealing with infinite geometric series or percentage decay.
Why: Only convergent series ($|r| < 1$) have a finite sum to infinity.

Example:

Problem: Find the sum to infinity of the geometric series with $a = 4$ and $r = \frac{1}{3}$.

Step 1: Check the condition Since $|r| = \frac{1}{3} < 1$, the series converges.

Step 2: Apply the formula

$$S_{\infty} = \frac{a}{1-r} = \frac{4}{1-\frac{1}{3}} = \frac{4}{\frac{2}{3}} = 6$$

Conclusion: The sum to infinity is **6**.

Strategy 6: Use Logs to Solve for Time

$$T = \frac{\log(A/P)}{\log(1+r/100)}$$

- What:** Use logarithms to solve for the time T in exponential growth or decay problems.
When: Use when the question asks, "How long until a value doubles/triples?"
Why: Logarithms are the inverse of exponentiation, allowing you to solve for the exponent T .

Example:

Problem: How long will it take for an investment of \$5000 to double at an interest rate of 6% per year?

Step 1: Identify variables - $P = 5000$, $A = 10000$, $r = 6$

Step 2: Apply the formula

$$T = \frac{\log\left(\frac{10000}{5000}\right)}{\log(1+0.06)} = \frac{\log(2)}{\log(1.06)}$$

$$T = \frac{0.3010}{0.0253} \approx 11.89$$

Conclusion: It takes **about 12 years** for the investment to double.

SECTION 2: COORDINATE GEOMETRY, VECTORS & TRIGONOMETRY

Coordinate Geometry — Straight Lines

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- What:** Gradient (slope) of a straight line through two points (x_1, y_1) and (x_2, y_2) .
When: Use whenever you need the steepness or direction of a line from two known points.
Why: Gradient measures the rate of change of y with respect to x .

$$y = mx + c \quad (\text{slope-intercept form})$$

$$y - y_1 = m(x - x_1) \quad (\text{point-gradient form})$$

$$ax + by + c = 0 \quad (\text{general form})$$

- What:** Three equivalent ways to write the equation of a straight line.
When: Use **point-gradient form** when you have a point and a gradient. Use **slope-intercept** when you need to read off m and c directly.
Why: All three forms represent the same line — choose the one that matches the given information.

$$\text{Parallel lines: } m_1 = m_2$$

$$\text{Perpendicular lines: } m_1 \times m_2 = -1$$

- What:** Conditions on gradients for parallel and perpendicular lines.
When: Use to check if two lines are parallel/perpendicular, or to find the gradient of a line perpendicular to a given one.
Why: Parallel lines have the same direction so same gradient. Perpendicular lines have gradients that are negative reciprocals of each other: $m_2 = -\frac{1}{m_1}$.

Point of intersection: solve the two line equations simultaneously

- What:** The point where two straight lines cross.
When: Use when asked to find where two lines meet — set them equal and solve.
Why: At the intersection point, both equations are satisfied simultaneously.

Coordinate Geometry — What, When, and Why

PART 1: Equation of a Circle

$$x^2 + y^2 + 2fx + 2gy + c = 0$$

- What:** The **general form** of the equation of a circle used in CSEC.
When: Use when the equation is given in this expanded format.
Why: It's a standard CSEC format and is useful for finding the centre and radius through formulas or completing the square.

Finding the Centre and Radius from General Form

From:

$$x^2 + y^2 + 2fx + 2gy + c = 0$$

We get:

- **Centre:**

$$(-f, -g)$$

- **Radius:**

$$r = \sqrt{f^2 + g^2 - c}$$

- When:** Use when you're given the general form and need to identify key features.
Why: These formulas are derived from completing the square and comparing to standard form.

Standard Form of a Circle

$$(x + f)^2 + (y + g)^2 = r^2$$

Why Use Each Form?

Form	Use When	Why
General Form	Given in exam	Easier to identify coefficients, complete the square, or use formulas
Standard Form	Centre and radius are known or needed	Makes substitution, sketching, and intersections easier

Converting General to Standard (Traditional)

Example: May/June 2017 Q9a

Problem: Convert

$$x^2 + y^2 + 4x - 6y - 3 = 0$$

to standard form. Find the centre and radius.

Step 1: Group x-terms and y-terms:

$$(x^2 + 4x) + (y^2 - 6y) = 3$$

Step 2: Complete the square:

For x -terms:

$$x^2 + 4x = (x + 2)^2 - 4$$

For y -terms:

$$y^2 - 6y = (y - 3)^2 - 9$$

Step 3: Substitute back into the equation:

$$(x + 2)^2 - 4 + (y - 3)^2 - 9 = 3$$

Step 4: Rearrange:

$$(x + 2)^2 + (y - 3)^2 - 13 = 3$$

$$(x + 2)^2 + (y - 3)^2 = 16$$

Conclusion: - Standard form:

$$(x + 2)^2 + (y - 3)^2 = 16$$

- Centre: $(-2, 3)$ - Radius: $\sqrt{16} = 4$

Method 2: Converting General to Standard (Easier)

Same Example (Alternate Method)

Problem: Convert

$$x^2 + y^2 + 4x - 6y - 3 = 0$$

to standard form using direct comparison.

Step 1: Compare to general form:

$$x^2 + y^2 + 2fx + 2gy + c = 0$$

So:

$$2f = 4 \Rightarrow f = 2$$

$$2g = -6 \Rightarrow g = -3$$

$$c = -3$$

Step 2: Use formulas:

- Centre:

$$(-f, -g) = (-2, 3)$$

- Radius:

$$r = \sqrt{f^2 + g^2 - c}$$

$$r = \sqrt{2^2 + (-3)^2 - (-3)}$$

$$r = \sqrt{4 + 9 + 3}$$

$$r = \sqrt{16} = 4$$

Step 3: Substitute into standard form:

$$(x + f)^2 + (y + g)^2 = r^2$$

$$(x + 2)^2 + (y - 3)^2 = 16$$

Conclusion: - Standard form:

$$(x + 2)^2 + (y - 3)^2 = 16$$

PART 2: Standard $\leftrightarrow$ General Form and Circle from Diameter

7. Converting Standard Form $\rightarrow$ General Form

Example: Start from

$$(x - 3)^2 + (y + 1)^2 = 25$$

Step 1: Expand both squared terms

$$x^2 - 6x + 9 + y^2 + 2y + 1 = 25$$

Step 2: Combine like terms

$$x^2 + y^2 - 6x + 2y + 10 = 25$$

Step 3: Bring all terms to one side

$$x^2 + y^2 - 6x + 2y + 10 - 25 = 0 \Rightarrow \boxed{x^2 + y^2 - 6x + 2y - 15 = 0}$$

What: General form of a circle.

When: Needed for comparison or matching to exam-style form.

Why: Expanding helps apply formulas like centre, radius, or intersection.

8. Equation of a Circle from Diameter Endpoints

Example: May/June 2015 Q9a(ii) The endpoints of the diameter of a circle are $A(-2, 1)$, $B(4, -3)$. Find the equation of the circle.

Step 1: Find the centre (midpoint)

$$\left(\frac{-2 + 4}{2}, \frac{1 + (-3)}{2} \right) = (1, -1)$$

Step 2: Find the radius (half of distance AB) Distance formula:

$$AB = \sqrt{(4 - (-2))^2 + (-3 - 1)^2} = \sqrt{6^2 + (-4)^2} = \sqrt{36 + 16} = \sqrt{52}$$

So:

$$r = \frac{1}{2}\sqrt{52} = \sqrt{13} \Rightarrow r^2 = 13$$

Step 3: Final Equation

$$\boxed{(x + 1)^2 + (y - 1)^2 = 13}$$

What: Equation of circle from diameter.

When: Given two endpoints of diameter.

Why: Midpoint = centre, half the distance = radius.

PART 3: Tangents, Normals, Midpoint, Distance, and Perpendicular Bisectors**9. Tangent to a Circle**

Formula: If the gradient of the radius is m_r , then:

$$m_{\text{tangent}} = -\frac{1}{m_r}$$

Then use point-gradient form to find the equation:

$$y - y_1 = m(x - x_1)$$

Example: May/June 2019 Q9b(i) The equation of a circle is:

$$x^2 + y^2 + 6x - 8y + 9 = 0$$

Find the equation of the tangent to the circle at the point $P(-3, 2)$.

Step 1: Compare to general form

$$x^2 + y^2 + 2fx + 2gy + c = 0 \Rightarrow 2f = 6 \Rightarrow f = 3, \quad 2g = -8 \Rightarrow g = -4$$

Centre: $(-f, -g) = (-3, 4)$

Step 2: Gradient of radius from centre to point $P(-3, 2)$

$$m_r = \frac{2 - 4}{-3 - (-3)} = \frac{-2}{0} \Rightarrow \text{Undefined}$$

Radius is vertical $\Rightarrow$ Tangent is horizontal.

Step 3: Tangent through $P(-3, 2)$ with horizontal gradient

$$\boxed{y = 2}$$

10. Normal to a Circle

Example: Worked version of May/June 2019 Q9b(ii) Using the same circle and point $P(-3, 2)$, find the equation of the normal.

Step 1: Centre: $(-3, 4)$, Point: $P(-3, 2)$ Gradient of normal:

$$m = \frac{2 - 4}{-3 - (-3)} = \frac{-2}{0} \Rightarrow \text{Undefined}$$

Normal is vertical $\Rightarrow$

$$\boxed{x = -3}$$

11. Midpoint

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

12. Distance

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

13. Perpendicular Bisector

- Steps:**
1. Find midpoint
 2. Find gradient of segment
 3. Use perpendicular gradient $m_{\perp} = -\frac{1}{m}$
 4. Use point-gradient form with the midpoint

Example: May/June 2016 Q9c Find the equation of the perpendicular bisector of the line joining $A(0, 2)$ and $B(4, -2)$.

Step 1: Midpoint

$$\left(\frac{0+4}{2}, \frac{2+(-2)}{2}\right) = (2, 0)$$

Step 2: Gradient of AB

$$m = \frac{-2-2}{4-0} = \frac{-4}{4} = -1$$

Step 3: Perpendicular gradient

$$m_{\perp} = -\frac{1}{-1} = 1$$

Step 4: Equation using point-gradient form

$$y - 0 = 1(x - 2) \Rightarrow \boxed{y = x - 2}$$

PART 4: Intersections, Loci, and Mixed Applications

14. Points of Intersection: Line and Circle

Method: To find where a line intersects a circle: 1. Substitute the linear equation into the circle equation

2. Rearrange into a quadratic equation in x or y

3. Solve the quadratic:

- 2 real roots $\rightarrow$ 2 points of intersection

- 1 root $\rightarrow$ tangent (1 point of contact)

- No real root $\rightarrow$ line does not intersect the circle

Example: May/June 2012 Q10b(i) Circle:

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

Line:

$$y = x - 2$$

Step 1: Substitute into the circle

$$x^2 + (x - 2)^2 - 4x + 6(x - 2) - 12 = 0$$

Step 2: Expand

$$x^2 + x^2 - 4x + 4 - 4x + 6x - 12 - 12 = 0$$

$$2x^2 - 2x - 20 = 0$$

Step 3: Solve the quadratic

$$x = \frac{1 \pm \sqrt{1^2 + 4(10)}}{2} = \frac{1 \pm \sqrt{41}}{2}$$

Step 4: Find corresponding y -values using $y = x - 2$

$$\left(\frac{1 + \sqrt{41}}{2}, \frac{-3 + \sqrt{41}}{2} \right) \quad \text{and} \quad \left(\frac{1 - \sqrt{41}}{2}, \frac{-3 - \sqrt{41}}{2} \right)$$

15. Locus Problems (Circle Condition)

Example: May/June 2023 Q10c A point moves such that its distance from the point $(1, 2)$ is always 5. Find the equation of its path.

Step 1: Centre = $(1, 2)$

Step 2: Radius = 5

$$(x - 1)^2 + (y - 2)^2 = 25$$

Final Answer:

$$(x - 1)^2 + (y - 2)^2 = 25$$

CSEC Exam Tips (Wrap-Up)

- Always convert the circle to standard form if possible.
- If the line is given as $y = mx + c$, substitute into the circle and solve the quadratic.
- The discriminant tells you how many points of intersection:
 - $D > 0$: 2 points
 - $D = 0$: tangent
 - $D < 0$: no intersection
- Use midpoint and distance frequently in diameter and bisector questions.
- Sketching the circle + line helps visualize intersections.

Vectors

A **vector** has both magnitude and direction. A **scalar** has magnitude only.

Vector Notation and Representation

$$\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix} = x\mathbf{i} + y\mathbf{j}$$

What: Two equivalent ways to write a 2D vector.

When: Use either form — both are acceptable in CSEC. Convert between them freely.

Why: $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are the unit vectors along the x - and y -axes.

$$\text{Equal vectors: } \mathbf{a} = \mathbf{b} \iff a_1 = b_1 \text{ and } a_2 = b_2$$

What: Two vectors are equal if and only if their corresponding components are equal.

When: Use when setting up equations from a vector equality, e.g. to find unknown values of x and y .

Why: Vectors encode both magnitude and direction — equality requires both to match, which means every component matches.

Magnitude (Modulus) of a Vector

$$|\mathbf{v}| = \sqrt{x^2 + y^2} \quad \text{where } \mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

What: The length (magnitude) of a vector.

When: Use when finding speed from a velocity vector, or the length of a displacement.

Why: Pythagoras' theorem applied to the horizontal and vertical components.

$$\text{Direction of } \mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}: \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

What: The angle θ that vector $\mathbf{v}$ makes with the positive x -axis.

When: Use when asked for the direction or bearing of a vector.

Why: $\tan \theta = \frac{\text{vertical component}}{\text{horizontal component}}$. Always check which quadrant the vector lies in to assign the correct angle.

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

What: A vector of magnitude 1 in the direction of $\mathbf{v}$.

When: Use when asked to find a unit vector or a vector of given magnitude in a specific direction.

Why: Dividing by the magnitude scales the vector to length 1 while preserving direction.

Vector Addition, Subtraction, and Scalar Multiplication

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}, \quad \mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \end{pmatrix}, \quad k\mathbf{a} = \begin{pmatrix} ka_1 \\ ka_2 \end{pmatrix}$$

What: Component-wise operations on vectors.

When: Use whenever you need to combine or scale vectors.

Why: Each component operates independently — treat x - and y -parts separately.

Position and Displacement Vectors

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} \quad (\text{displacement from } A \text{ to } B)$$

What: The vector from point A to point B , given their position vectors $\mathbf{a}$ and $\mathbf{b}$.

When: Use when finding the direction or distance between two points using vectors.

Why: $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$. Think: “end minus start”.

Scalar (Dot) Product

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 = |\mathbf{a}||\mathbf{b}| \cos \theta$$

What: Two equivalent definitions of the scalar (dot) product.

When: Use to find the angle between two vectors, or to test for perpendicularity.

Why: The two forms let you calculate the dot product from components *or* from magnitudes and the angle between the vectors.

Angle Between Two Vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

- What:** Formula for the angle θ between vectors $\mathbf{a}$ and $\mathbf{b}$.
When: Use when a question asks for “the angle between” two vectors.
Why: Rearranged from the dot product definition. Always gives $0 \leq \theta \leq 180^\circ$.

Parallel and Perpendicular Vectors

Perpendicular: $\mathbf{a} \cdot \mathbf{b} = 0$

Parallel: $\mathbf{a} = k\mathbf{b}$ for some scalar k

- What:** Conditions for two vectors to be perpendicular or parallel.
When: Use to check or prove that two lines/vectors are perpendicular/parallel.
Why: Perpendicular: $\cos 90^\circ = 0$, so $\mathbf{a} \cdot \mathbf{b} = 0$. Parallel: one vector is a scalar multiple of the other.

Example: Finding the Angle Between Two Vectors

Let $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

$$\mathbf{a} \cdot \mathbf{b} = (3)(1) + (4)(-2) = 3 - 8 = -5$$

$$|\mathbf{a}| = \sqrt{9 + 16} = 5, \quad |\mathbf{b}| = \sqrt{1 + 4} = \sqrt{5}$$

$$\cos \theta = \frac{-5}{5\sqrt{5}} = \frac{-1}{\sqrt{5}} \Rightarrow \theta = \cos^{-1}\left(\frac{-1}{\sqrt{5}}\right) \approx 116.6^\circ$$

Trigonometry

Radians and Arc/Sector Formulae

$$\pi \text{ radians} = 180^\circ \quad \Rightarrow \quad 1 \text{ rad} = \frac{180^\circ}{\pi}, \quad 1^\circ = \frac{\pi}{180} \text{ rad}$$

What: Conversion between degrees and radians.

When: Use whenever the angle is given/required in radians (most CSEC trig questions specify radians).

Why: Radians are the natural unit for angle in calculus and circular measure.

$$\text{Arc Length: } \ell = r\theta \quad \text{Sector Area: } A = \frac{1}{2}r^2\theta$$

What: Formulae for a sector of a circle with radius r and angle θ in **radians**.

When: Use in any question involving an arc, sector, or segment of a circle.

Why: These come directly from the fraction $\frac{\theta}{2\pi}$ of the full circle's circumference/area.

Area of a segment (region between a chord and arc):

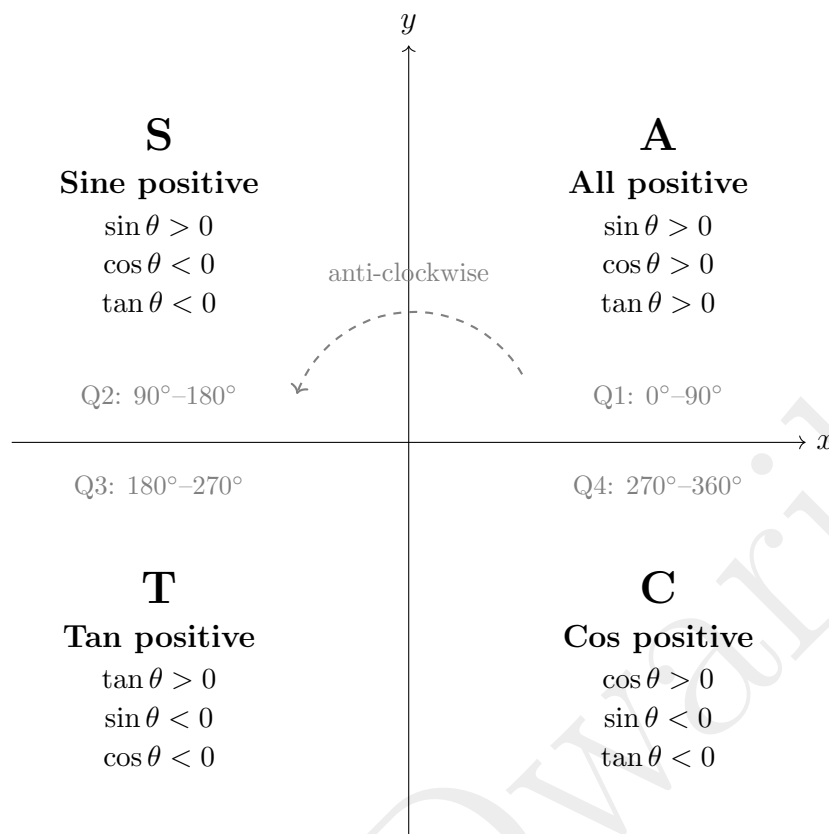
$$A_{\text{segment}} = \frac{1}{2}r^2(\theta - \sin \theta)$$

Exact Values Table

Angle	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
Degrees	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.

CAST Diagram — Signs in Each Quadrant

Mnemonic: “All School Teachers Cuss” — going **anti-clockwise** from Q1:



Q1	Q2	Q3	Q4
All	School	Teachers	Cuss
All +	sin +	tan +	cos +
0°–90°	90°–180°	180°–270°	270°–360°

What: Shows which trig ratios are positive in each quadrant.

When: Use when finding all solutions to a trig equation in $[0^\circ, 360^\circ]$ or $[0, 2\pi]$.

Why: The mnemonic **All School Teachers Cuss** tells you, going anti-clockwise from Q1, which ratio is positive in each quadrant. Once you find the principal (acute) angle, CAST tells you which other quadrant gives a valid solution.

Related Angle Rules

$$\text{Q2: } \sin(180^\circ - \theta) = \sin \theta, \quad \cos(180^\circ - \theta) = -\cos \theta$$

$$\text{Q3: } \sin(180^\circ + \theta) = -\sin \theta, \quad \cos(180^\circ + \theta) = -\cos \theta$$

$$\text{Q4: } \sin(360^\circ - \theta) = -\sin \theta, \quad \cos(360^\circ - \theta) = \cos \theta$$

Pythagorean and Quotient Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

- What:** The fundamental Pythagorean identity.
When: Use to prove identities or simplify/substitute expressions involving $\sin^2$ and $\cos^2$.
Why: Comes directly from Pythagoras' theorem on the unit circle.

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

- What:** Quotient identity — defines tangent in terms of sine and cosine.
When: Use when simplifying expressions containing $\tan \theta$, or to eliminate $\tan$ from equations.
Why: This definition follows directly from the unit circle.

Derived forms of the Pythagorean identity (useful for substitution):

$$\sin^2 \theta = 1 - \cos^2 \theta \quad \cos^2 \theta = 1 - \sin^2 \theta$$

Compound Angle Formulae

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

- What:** Formulae for the trig ratios of the sum or difference of two angles.
When: Use to expand $\sin(75^\circ)$, simplify $\cos(A - B)$, or prove identities.
Why: These are given on the CSEC formula sheet but you must know how to apply them. Note the $\mp$ sign pattern in $\cos$ — it is the **opposite** of the $\pm$.

Double Angle Formulae

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

- What:** Special case of compound angle formulae where $B = A$.
- When:** Use to simplify $\sin 2x$, solve equations involving $\cos 2x$, or prove identities.
- Why:** $\cos 2A$ has 3 equivalent forms — choose the one that matches the other terms in the problem to simplify most efficiently.

Choosing the Right Form of $\cos 2A$

Use this form	When you see...
$1 - 2\sin^2 A$	other $\sin^2$ terms in the equation
$2\cos^2 A - 1$	other $\cos^2$ terms in the equation
$\cos^2 A - \sin^2 A$	a mix of both, or an identity proof

Solving Trigonometric Equations

General Strategy

1. Rearrange to get a single trig function on one side (e.g. $\sin \theta = k$).
2. Find the principal (acute) angle: $\alpha = \sin^{-1}(|k|)$.
3. Use CAST to find all solutions in the given range.
4. For equations involving $\sin 2\theta$, substitute $\phi = 2\theta$ and extend the range accordingly.

Example: Solve $\cos 2\theta = 1 - 2\sin^2 \theta + \sin \theta - 1$ is typical — substitute $\cos 2\theta = 1 - 2\sin^2 \theta$ to get a quadratic in $\sin \theta$.

Area of a Triangle

$$\text{Area} = \frac{1}{2}ab \sin C$$

- What:** Area of any triangle given two sides a , b and the included angle C between them.
- When:** Use when two sides and the angle between them are known (as opposed to base and perpendicular height).
- Why:** The perpendicular height of the triangle equals $a \sin C$, so $\text{Area} = \frac{1}{2} \times b \times a \sin C$.

Graphs of Trigonometric Functions

Function	Amplitude	Period	Key feature
$\sin kx$	1	$\frac{2\pi}{k}$	Starts at 0, rises first
$\cos kx$	1	$\frac{2\pi}{k}$	Starts at 1 (maximum)
$\tan kx$	undefined	$\frac{\pi}{k}$	Has vertical asymptotes

What: Key properties of the basic trig graphs for $k = 1$ or $k = 2$.

When: Use when asked to sketch or describe the graph of $\sin kx$, $\cos kx$, or $\tan kx$.

Why: The factor k compresses the graph horizontally — doubling k halves the period, doubling the number of cycles in $[0, 2\pi]$.

Key values to plot for $\sin x$ and $\cos x$:

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin x$	0	1	0	-1	0
$\cos x$	1	0	-1	0	1

SECTION 3: INTRODUCTORY CALCULUS

Differentiation

Note: Differentiation is the process of finding the **gradient** (or rate of change) of a curve.

The symbol $\frac{dy}{dx}$ means “change in y over change in x ” — the same concept as the formula for gradient from coordinate geometry:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{corresponds to} \quad \frac{dy}{dx} = \text{gradient at a point}$$

So while $m = \frac{y_2 - y_1}{x_2 - x_1}$ gives the *average* gradient between two points, $\frac{dy}{dx}$ gives the *instantaneous* gradient at a specific point on a curve.

Basic Differentiation Rules

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

What: Power Rule — Used to differentiate powers of x .
When: Anytime you're asked to find $\frac{dy}{dx}$ for expressions like x^3 , $x^{\frac{1}{2}}$, etc.
Why: It gives the slope at any point on a polynomial curve.

$$\frac{d}{dx}(c) = 0$$

What: Constant Rule.
When: Used when differentiating constants (e.g. 4, -7, π).
Why: Constants have zero rate of change — their graphs are flat.

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

What: Sum and difference rule.
When: Used when differentiating multiple terms added or subtracted.
Why: You can differentiate each part separately.

$$\frac{d}{dx}(k \cdot f(x)) = k \cdot \frac{d}{dx}(f(x))$$

- What:** Constant multiple rule.
When: Used when a number multiplies a function (e.g. $3x^2$).
Why: The constant factor remains and the rest differentiates normally.

Example: Differentiate

$$y = 4x^3 - 5x^2 + 7x - 9$$

$$\frac{dy}{dx} = \frac{d}{dx}(4x^3) - \frac{d}{dx}(5x^2) + \frac{d}{dx}(7x) - \frac{d}{dx}(9) = 12x^2 - 10x + 7$$

Final Answer: $\frac{dy}{dx} = 12x^2 - 10x + 7$

$$\frac{d}{dx}(\sin x) = \cos x \quad \frac{d}{dx}(\cos x) = -\sin x$$

- What:** Basic trigonometric differentiation rules.
When: Used when differentiating $\sin x$ and $\cos x$.
Why: These are standard identities for the slope of sine and cosine curves.

$$\frac{d}{dx}(\tan x) = \sec^2 x \quad (\text{Valid for } x \neq \frac{\pi}{2} + n\pi)$$

- What:** Derivative of tangent.
When: Used for curves involving $\tan x$, where it's defined.
Why: This gives the slope of $\tan x$, which increases rapidly near asymptotes.

$$\frac{d}{dx}[u(x) \cdot v(x)] = u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx}$$

- What:** Product Rule.
When: Use when differentiating the product of two functions.
Why: You must apply this rule since the derivative of a product is not simply the product of derivatives.

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2}$$

- What:** Quotient Rule.
When: Use when differentiating the division of two functions.
Why: This rule accounts for how the top and bottom both change with respect to x .

$$\frac{d}{dx}[f(ax + b)] = a \cdot f'(ax + b)$$

What: Chain rule shortcut for linear inner functions.

When: Use when differentiating functions like $\sin(3x)$, $(5x + 1)^4$, etc.

Why: The derivative of the inside is multiplied by the derivative of the outside.

Examples Using Chain Rule Shortcut

Example 1: Differentiate $y = \sin(3x)$

$$\begin{aligned}\frac{dy}{dx} &= \cos(3x) \cdot \frac{d}{dx}(3x) \\ &= \cos(3x) \cdot 3 \\ &= \boxed{3 \cos(3x)}\end{aligned}$$

Example 2: Differentiate $y = (5x + 1)^4$

$$\begin{aligned}\frac{dy}{dx} &= 4(5x + 1)^3 \cdot \frac{d}{dx}(5x + 1) \\ &= 4(5x + 1)^3 \cdot 5 \\ &= \boxed{20(5x + 1)^3}\end{aligned}$$

Second Derivative

$$\frac{d^2y}{dx^2} = f''(x) = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

What: The second derivative — differentiating $\frac{dy}{dx}$ a second time.

When: Use to determine the nature of stationary points, or when asked for $f''(x)$.

Why: The second derivative measures the rate of change of the gradient itself — it tells you whether the curve is concave up or down.

Stationary Points

$$\text{Stationary point occurs when } \frac{dy}{dx} = 0$$

- What:** Points on a curve where the gradient is zero (the tangent is horizontal).
When: Use to find maximum, minimum, or turning points of a function.
Why: At a turning point the curve momentarily stops increasing or decreasing, so its gradient is zero.

Nature of Stationary Points — Two Methods

Method 1: Second Derivative Test

$$f''(x) > 0 \Rightarrow \text{Minimum} \quad f''(x) < 0 \Rightarrow \text{Maximum} \quad f''(x) = 0 \Rightarrow \text{Test inconclusive}$$

- What:** Quick test using the second derivative to classify a stationary point.
When: Use first — it's faster. If $f''(x) = 0$, fall back to Method 2.
Why: $f'' > 0$ means the gradient is increasing (curve curves upward = minimum). $f'' < 0$ means gradient is decreasing (curves downward = maximum).

Method 2: First Derivative Sign Change Test

Sign of $f'(x)$	−	0	+	+	0	−
Nature				Minimum		Maximum

- What:** Check the sign of $f'(x)$ either side of the stationary point.
When: Use when the second derivative is zero or hard to compute.
Why: If the gradient goes from negative to positive, the curve dips then rises — a minimum. If positive to negative, it peaks — a maximum.

Tangent and Normal to a Curve

$$\text{Gradient of tangent at } x = a: m_T = f'(a)$$

$$\text{Gradient of normal at } x = a: m_N = -\frac{1}{f'(a)}$$

$$\text{Equation (use point-gradient form): } y - y_1 = m(x - x_1)$$

- What:** The tangent touches the curve at one point; the normal is perpendicular to the tangent at the same point.
When: Use when asked for the equation of the tangent or normal to a curve $y = f(x)$ at a given point.
Why: The derivative gives the gradient of the tangent directly. The normal is perpendicular, so its gradient is the negative reciprocal.

Example

Find the equation of the tangent to $y = x^3 - 2x$ at $x = 2$.

$$f'(x) = 3x^2 - 2 \Rightarrow f'(2) = 12 - 2 = 10 \quad (= m_T)$$

$$y\text{-value: } f(2) = 8 - 4 = 4 \Rightarrow \text{point } (2, 4)$$

$$\text{Tangent: } y - 4 = 10(x - 2) \Rightarrow \boxed{y = 10x - 16}$$

Rate of Change Applications

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad (\text{Chain Rule for related rates})$$

What: Connects the rate of change of two related variables through a third.

When: Use when asked how fast area, volume, or another quantity is changing with time, given a rate like $\frac{dr}{dt}$.

Why: If y depends on u , and u depends on x , the overall rate is the product of the individual rates.

Example: Area of circle $A = \pi r^2$, so $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} = 2\pi r \cdot \frac{dr}{dt}$.

Differentiation from First Principles

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

What: This is the definition of the derivative using first principles.

When: Use when asked to find the derivative using its definition.

Why: It gives the slope of the tangent by finding the limit of the slope of a secant line as the interval becomes infinitely small.

Note: This formula expresses the **instantaneous rate of change** as the limit of the *average rate of change*. It connects directly to the gradient formula from coordinate geometry:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \longrightarrow \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Integration

Note: Integration is the reverse (or “undoing”) of differentiation.

It answers questions like: “What function gave rise to this derivative?” or “What is the total accumulated change?” You can use integration to:

- Recover the original function from its derivative

- Find the area under a curve between two points
- Find total distance travelled or volume (with special methods)

Just as $\frac{dy}{dx}$ measures how fast y is changing, integration adds up all those little changes to reconstruct y — or to measure total effect (like area).

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad \text{for } n \neq -1$$

What: Power rule for integration.
When: Used to integrate any power of x , as long as the exponent is not -1 .
Why: It reverses differentiation using the rule $\frac{d}{dx}(x^n) = nx^{n-1}$.

$$\int k dx = kx + c$$

What: Integration of a constant.
When: Use when you're integrating a fixed number like 3, -7, or π .
Why: A constant integrates to a linear expression because its derivative would be zero.

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

What: Sum and difference rule.
When: When you're integrating multiple terms added or subtracted.
Why: Each term can be integrated separately — integration is linear.

$$\int k \cdot f(x) dx = k \cdot \int f(x) dx$$

What: Constant multiple rule.
When: Use when a constant is multiplied by a function.
Why: The constant “comes outside” the integral — just like in differentiation.

$$\int \sin x dx = -\cos x + c \quad \int \cos x dx = \sin x + c$$

What: Trigonometric integration rules.
When: Use when integrating $\sin x$ or $\cos x$.
Why: These reverse the basic derivatives: $\frac{d}{dx}(\cos x) = -\sin x$, etc.

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax) + c \quad \int \cos(ax) dx = \frac{1}{a} \sin(ax) + c$$

What: Trigonometric integration using chain rule shortcut.
When: Use when integrating $\sin(ax)$ or $\cos(ax)$ where a is a number.
Why: The derivative of the inside is divided out after integrating.

$$\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + c \quad \text{for } n \neq -1$$

What: Power rule with a linear term inside.
When: Use when integrating functions like $(3x + 1)^4$ or $(2x - 5)^5$.
Why: You apply the reverse power rule and then divide by the coefficient of x .

$$\int \sec^2 x dx = \tan x + c$$

What: Reverse of the derivative of $\tan x$.
When: Use when the integrand is $\sec^2 x$.
Why: Because $\frac{d}{dx}(\tan x) = \sec^2 x$.

$$\int \frac{1}{x} dx = \ln |x| + c$$

What: Special case not covered by the power rule.
When: Use when the integrand is exactly $\frac{1}{x}$.
Why: Because this is the reverse of $\frac{d}{dx}(\ln |x|) = \frac{1}{x}$.

Applications of Integration: Area and Volume

$$\text{Area under a curve: } A = \int_a^b f(x) dx$$

What: This formula calculates the exact area between the curve $y = f(x)$, the x -axis, and vertical lines $x = a$ and $x = b$.
When: Use when asked to find the area between a curve and the x -axis, or between a curve and a horizontal line.
Why: Integration adds up infinitely thin slices (rectangles) under the curve to give total area.

Volume of solid formed by rotating a curve about the x-axis: $V = \pi \int_a^b [f(x)]^2 dx$

- What:** This formula gives the volume of the solid formed when the curve $y = f(x)$ is rotated 360° about the x-axis.
- When:** Use when asked to find the volume generated by rotating a region between $x = a$ and $x = b$.
- Why:** Each horizontal slice becomes a disc, and integration adds up the volume of all the discs.

Equation of a Curve from its Gradient Function

If $\frac{dy}{dx} = f(x)$ and the curve passes through (x_1, y_1) , then:

$$y = \int f(x) dx + c \quad \text{where } c \text{ is found by substituting } (x_1, y_1)$$

- What:** Finding the equation of a curve when its derivative and one point are known.
- When:** Use when a question gives you $\frac{dy}{dx}$ and a point, and asks for y .
- Why:** Integration reverses differentiation but produces a constant c . The given point provides the extra information needed to find c .

Example

Given $\frac{dy}{dx} = 3x^2 - 4$ and the curve passes through $(2, 5)$, find y .

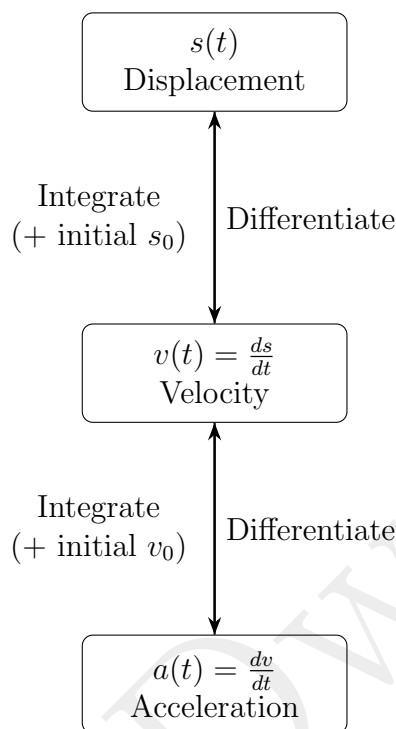
$$y = \int (3x^2 - 4) dx = x^3 - 4x + c$$

Substitute $(2, 5)$: $5 = 8 - 8 + c \Rightarrow c = 5$

$$y = x^3 - 4x + 5$$

Kinematics — Differentiation and Integration Flow

The kinematics topic is a direct application of both **differentiation** and **integration**:



Interpretation:

- Differentiating $s(t)$ gives velocity, and differentiating $v(t)$ gives acceleration.
- To move in reverse (from acceleration to velocity or displacement), integrate — but **include initial conditions** like $v(0)$ or $s(0)$.

Equations of Motion

$$v = \frac{ds}{dt} \quad a = \frac{dv}{dt}$$

What: Definitions of velocity and acceleration as derivatives.

When: Use when given a function for displacement or velocity and asked to find instantaneous values.

Why: These directly apply differentiation to kinematics.

$$s = \int v \, dt \quad v = \int a \, dt$$

What: Formulas to recover displacement or velocity using integration.
When: Use when given acceleration or velocity functions and initial values.
Why: These are reverse processes of differentiation and require initial conditions.

$$v = u + at$$

What: First SUVAT equation — final velocity.
When: Use when acceleration is constant.
Why: Derived from integrating acceleration.

$$s = ut + \frac{1}{2}at^2$$

What: Displacement with constant acceleration.
When: Use when you know initial velocity and acceleration.
Why: Derived by integrating $v = u + at$.

$$v^2 = u^2 + 2as$$

What: SUVAT equation without time.
When: Use when time t is not given.
Why: Derived by eliminating t from earlier SUVAT equations.

Displacement vs Total Distance Travelled

$$\text{Displacement} = \int_a^b v(t) dt \quad (\text{can be negative if moving backwards})$$

$$\text{Total Distance} = \int_a^b |v(t)| dt \quad (\text{always positive})$$

What: The difference between displacement (net change in position) and total distance (total path length).
When: When the particle **changes direction** during the interval, the definite integral of v gives displacement only. To find total distance, split at the point(s) where $v = 0$ and add the absolute values of each part.
Why: When $v < 0$ the particle moves backwards — the integral subtracts that portion. Total distance counts all movement, so use $|v|$.

Strategy for Total Distance

1. Find when $v(t) = 0$ — these are the times when the particle changes direction.

2. Integrate $v(t)$ over each sub-interval separately.
3. Add the **absolute values** of each result.

SECTION 4: PROBABILITY & STATISTICS

Statistics

Types of Data

Type	Meaning	Example
Qualitative	Non-numerical categories	Colours, gender
Quantitative — Discrete	Countable numbers	Number of students
Quantitative — Continuous	Any value in a range	Height, weight, time

Measures of Central Tendency

$$\bar{x} = \frac{\sum x_i}{n} \quad (\text{Ungrouped data mean})$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad (\text{Grouped/frequency data mean})$$

- What:** Arithmetic mean of ungrouped data, and of data presented in a frequency table.
- When:** Use the grouped form whenever data is given with frequencies or class intervals (use midpoint of each class).
- Why:** The mean balances all values — each $f_i x_i$ weights each midpoint x_i by its frequency.

Median: The middle value when data is arranged in order.

- Odd n : median is the $\left(\frac{n+1}{2}\right)$ th value.
- Even n : median is the average of the $\frac{n}{2}$ th and $\left(\frac{n}{2} + 1\right)$ th values.

Mode: The value (or class) that occurs most frequently.

Quartiles, Percentiles, and Box-and-Whisker Plots

Q_1	Q_2	Q_3
Lower Quartile	Median	Upper Quartile
25th percentile	50th percentile	75th percentile

- What:** Quartiles divide ordered data into four equal parts.
- When:** Use when asked for spread, to draw a box-and-whisker plot, or to find the interquartile range.
- Why:** They summarise the distribution of the data, especially the middle 50%.

Method 1 — Position using n (for grouped/large data):

$$Q_1 = \frac{1}{4}nth \text{ term} \quad Q_2 = \frac{1}{2}nth \text{ term} \quad Q_3 = \frac{3}{4}nth \text{ term}$$

Method 2 — Position using $(n + 1)$ (commonly taught for raw data):

$$Q_1 = \frac{1}{4}(n + 1)\text{th term} \quad Q_2 = \frac{1}{2}(n + 1)\text{th term} \quad Q_3 = \frac{3}{4}(n + 1)\text{th term}$$

General Percentile Position:

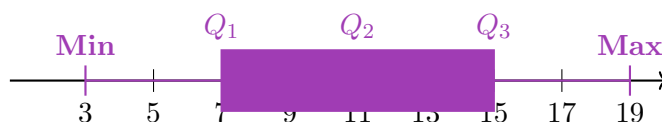
$$P_p = \frac{p}{100} \times nth \text{ term}$$

- What:** Two methods for locating quartile positions in an ordered dataset of n values.
- When:** Method 2 $(n + 1)$ is what we use for raw data in class. Method 1 (n) is used for grouped data or cumulative frequency. Always arrange data in ascending order first.
- Why:** If the position is not a whole number, round up to the next value; if it is a whole number, average that value and the next. Be consistent within a question.

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1 \quad \text{Semi-IQR} = \frac{Q_3 - Q_1}{2}$$

- What:** IQR measures the spread of the middle 50% of data. Semi-IQR is half of that.
- When:** Use as a measure of spread (especially when data is skewed or has outliers).
- Why:** Unlike range, IQR ignores extreme values and focuses on the central spread.

Box-and-Whisker Plot — displays the five-number summary: Minimum, Q_1 , Median (Q_2), Q_3 , Maximum.



- What:** The box spans Q_1 to Q_3 (the middle 50% of data). The line inside is the median. The whiskers extend to the minimum and maximum values.
- When:** Use to display the spread and skewness of a dataset at a glance.
- Why:** If the median line is closer to Q_1 , the data is positively skewed. If closer to Q_3 , it is negatively skewed. A centred line means roughly symmetric data.

Variance and Standard Deviation

$$\text{Variance (ungrouped): } S^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x_i^2}{n} - \bar{x}^2$$

- What:** The average of the squared deviations from the mean.
- When:** Use when asked for variance or standard deviation of a list of values.
- Why:** Squaring the differences ensures negatives don't cancel — it measures overall spread. The shortcut form $\frac{\sum x_i^2}{n} - \bar{x}^2$ avoids subtracting the mean from each value individually.

$$\text{Variance (grouped): } S^2 = \frac{\sum f_i(x_i - \bar{x})^2}{\sum f_i} = \frac{\sum f_i x_i^2}{\sum f_i} - \bar{x}^2$$

- What:** Variance formula for data given in a frequency table. Use midpoints x_i for class intervals.
- When:** Use when data is in a grouped frequency table.
- Why:** Same logic as ungrouped — each squared deviation is weighted by its frequency.

$$\text{Standard Deviation: } S = \sqrt{S^2}$$

- What:** The square root of variance — same units as the original data.
- When:** Use whenever a question asks for standard deviation. Always find variance first.
- Why:** Standard deviation is more interpretable than variance since it's in the same units as the data.

Step-by-Step: Variance and Standard Deviation (Ungrouped)

Data: 4, 7, 8, 9, 12. $n = 5$

$$\bar{x} = \frac{4 + 7 + 8 + 9 + 12}{5} = \frac{40}{5} = 8$$

$$S^2 = \frac{(4 - 8)^2 + (7 - 8)^2 + (8 - 8)^2 + (9 - 8)^2 + (12 - 8)^2}{5} = \frac{16 + 1 + 0 + 1 + 16}{5} = \frac{34}{5} = 6.8$$

$$S = \sqrt{6.8} \approx 2.61$$

Stem-and-Leaf Diagrams

A **stem-and-leaf diagram** displays raw data by splitting each value into a *stem* (leading digit(s)) and a *leaf* (final digit). It preserves individual data values and allows you to read off median, quartiles, and mode directly.

Example: Data: 12, 15, 23, 27, 31, 34, 34, 38

Stem	Leaf
1	2 5
2	3 7
3	1 4 4 8

Key: 1|2 means 12.

Probability

Key Definitions

Term	Meaning
Experiment	A process that produces an observable outcome
Outcome	A single result of an experiment
Sample Space S	The set of all possible outcomes
Event A	A subset of the sample space

Classical Probability

$$P(A) = \frac{\text{number of outcomes in } A}{\text{total number of outcomes in } S} \quad (\text{when all outcomes are equally likely})$$

- What:** The probability of event A occurring.
When: Use when outcomes are equally likely (coins, dice, cards, balls in a bag).
Why: We count favourable outcomes and divide by total possible outcomes.

Basic Laws of Probability

$$0 \leq P(A) \leq 1 \quad \sum_{\text{all outcomes}} P = 1 \quad P(A') = 1 - P(A)$$

- What:** The three fundamental axioms of probability.
When: Use $P(A') = 1 - P(A)$ when finding “the probability that A does NOT happen” is easier.
Why: A and its complement A' together cover the entire sample space, so their probabilities sum to 1.

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- What:** Probability that A **or** B occurs (or both).
When: Use for any two events — this formula is always valid.
Why: Adding $P(A)$ and $P(B)$ counts $P(A \cap B)$ twice, so we subtract it once.

Mutually Exclusive Events

$$\text{If } A \text{ and } B \text{ are mutually exclusive: } P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B)$$

- What:** Events that **cannot both occur** at the same time.
When: Use when events share no outcomes (e.g. rolling a 2 and rolling a 5 on one die).
Why: If they can't overlap, $P(A \cap B) = 0$, simplifying the addition rule.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{the probability of } A \text{ given that } B \text{ has occurred})$$

- What:** The updated probability of A given that we know B has already happened.
When: Use when information about one event changes the probability of another.
Why: Knowing B occurred restricts the sample space to B — we recalculate within that restricted space.

Independent Events

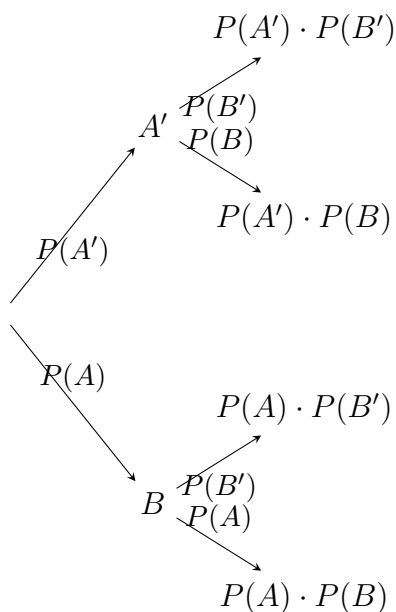
$$\text{If } A \text{ and } B \text{ are independent: } P(A \cap B) = P(A) \times P(B) \quad \text{and} \quad P(A|B) = P(A)$$

- What:** Events where the occurrence of one does **not** affect the probability of the other.
When: Use when events are from separate experiments (e.g. tossing two different coins).
Why: If B gives no information about A , then $P(A|B) = P(A)$, and multiplying the probabilities gives the joint probability.

Probability Diagrams

Tree Diagrams

Tree diagrams show multi-stage experiments. **Multiply along branches (AND), add across branches (OR).**



- What:** A branching diagram showing all possible outcomes of a multi-step experiment.
- When:** Use for two-stage experiments or sequential events (limited to two initial branches on CSEC).
- Why:** Multiplying along branches gives joint probabilities; summing branches with the same outcome gives total probability.

Venn Diagrams (Two Sets)

For two events A and B :

- **Inside both circles** $\Rightarrow A \cap B$ (A and B)
- **Inside A only** $\Rightarrow A \cap B'$
- **Inside B only** $\Rightarrow A' \cap B$
- **Outside both** $\Rightarrow A' \cap B' = (A \cup B)'$

$$P(A \text{ only}) = P(A) - P(A \cap B) \quad P(A \cup B)' = 1 - P(A \cup B)$$

- What:** Key Venn diagram probability expressions.
- When:** Use to fill in Venn diagram regions from given probabilities, then answer follow-up questions.
- Why:** All regions of the Venn diagram must sum to 1 (the entire sample space). Always fill in $A \cap B$ first.

Strategy: Filling a Venn Diagram

1. Fill in $P(A \cap B)$ first (the overlap).
2. Subtract the overlap from $P(A)$ and $P(B)$ to get the exclusive regions.
3. The region outside both $= 1 - P(A \cup B)$.

Possibility Space Diagrams

A grid showing all possible pairs of outcomes from two experiments (e.g. two dice). Count the squares satisfying a condition to find probability.

$$P(\text{event}) = \frac{\text{number of favourable squares in grid}}{\text{total squares in grid}}$$

Summary: Probability Laws

Law	Formula
Complement	$P(A') = 1 - P(A)$
Addition (general)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive	$P(A \cup B) = P(A) + P(B)$
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Independent	$P(A \cap B) = P(A) \times P(B)$